Hein Zarni Aung

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EDUCATION

Stony Brook University **Expected Graduation: May 2027**

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Full Stack Development, Systems Fundamentals I & II, Algorithms

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University **Sep 2025 – Dec 2025**

- Built a Python pipeline using Selenium to automate certificate and domain scraping for a real-time SSL/TLS fraud-detection system, then applied keyword- and heuristic-based filtering logic to flag sites impersonating EZ-Pass tolling services.

INTERNSHIPS

Software Engineering Intern — Clustr **Jan 2026 – Sept 2026 (expected)**

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC **Jun 2026 – Sept 2026 (expected)**

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Built a seed-data workflow and shared dev environment setup that let the rest of the team build and test against Jamory's database independently, removing a bottleneck that would've otherwise gated backend work to one person.

PROJECTS

TuneAcademy **React, NumPy, Essentia, Firestore**

- Built the audio-analysis backend pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy TuneAcademy to Cloudflare Workers.

Discord Voice Channel Activity Monitor **Node.js, Discord.js, Cloud, PM2, AWS**

- Built a Node.js/Discord.js bot to solve a scheduling problem for a friend group spread across different schedules, notifying users in real time when someone joins a voice channel so they can catch impromptu hangouts or gaming sessions.
- Implemented customizable DM notifications, VIP lists, and per-server/per-user channel thresholds with JSON-based configuration; deployed on a cloud server using PM2, reaching 50 concurrent active users.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University **Aug 2025 – Dec 2025**

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

Frontend: React, React Native, Next.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, Selenium, GitHub Actions